

**ADMINISTRATIVE INFORMATION****FMI No: 25399****FMI Title: LightSpeed 2X - 3X Slip Ring to OBC Wiring Error**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS – Americas	GEMS - Europe	GEMS – Asia
GE Medical Systems Insite Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

ATTENTION: Follow the procedure in this FMI instruction.

If the gantry is found to be mis-wired and/or mis-labeled, correct the error and close the FMI Service Dispatch as a Code 1 (FMI Completed).

If the gantry is wired and labeled properly, close the FMI Service Dispatch as a Code 2 (FMI Not Required)

Note: This FMI does not apply to CT/PET Systems

Unit Tracked: Refer to Effectivity list on page 4.

FMI Instructions Part No.: 5224728-1EN, Rev 3

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas

GEMS - Europe

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25399

FMI Title: LightSpeed 2X – 3X Slip Ring to OBC Wiring Error

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer

Comments

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Revision History

Revision	Date	Reason for Change
1	April 2 nd 2007	Initial release
2	May 21 st 2007	Milestone F3 feedback changes
3	June 14 th 2007	Pilot Feedback

Effectivity

The effectivity for this FMI applies to a select group of the CT LightSpeed systems manufactured in Hino, Japan between June 1st 2002 and December 31st 2002, which have H2 and H3 CT gantries (LightSpeed 2X-3X). The models tracked in effectivity are below:

Product	Gantry Assembly
GANTRY COLLECTOR 8 SLICE (LS Ultra)	2281177-3
GANTRY COLLECTOR 4 SLICE (LS Plus)	2281177-4
GANTRY COLLECTOR 4 SLICE (PS, HS QX/i)	2341799

Note: This FMI does not apply to CT/PET Systems

Purpose

Customer Impact

1. CT/PET iTrak Fixes

- a. PSR 13085844 - CT Gantry Mis-Wire at Univ of WI, Madison, WI
- b. PSR 13099568 - 120VAC connections reversed from slip ring to OBC

2. Problem: Possible "120V_{AC} Connections Reversed from Slip Ring to OBC"

The main issue raised by these reports, requiring a FMI, was for the possibility that CT LightSpeed 2X – 3X gantries may contain a wiring error, which reverses the 120 volt hot and neutral lines between the Slip Ring and the OBC chassis. The mis-labeling of the two wires by the Supplier caused this mis-wire in the gantry manufacturing process. The mis-wire error alone does not cause direct faults with the system, but indirectly can cause damage to gantry electrical components (mainly OBC Power I/F Board) if a short circuit occurs on the 120 volt hot line. With the 120 volt hot and neutral lines reversed, the hot line is no longer fuse protected, allowing 40 amps to flow through the OBC Power I/F Board traces.

Part Affected	Part Number
Harness – 120VAC S/R to OBC	2262943

Prerequisite FMIs

None.

Related FMIs

None

Furnished Materials

FMI Kit 5224728 consists of the following materials:

Item	Part No	Qty	Description
1	5224728-1EN	1	FMI 25399 Document
2	5224857	1	Labels for cable

Tool Requirements

Quantity	Description
	FE Toolkit
1	Philips #2 Screwdriver
1	Flat-blade Screwdriver
1	Metric Hex Key Set
1	ESD wrist-band
1	Multimeter (DVM)

Personnel Requirements

Resource	Labor	Travel
1 FE	1.0 hrs	1.0 hrs

Before You Begin

Before beginning this procedure, please read the gantry safety information.

Safety



⚠ WARNING

POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES WHILE PERFORMING FMI.

Procedure

Removing Gantry and Slip Ring Covers

1. Remove the Gantry side covers.



NOTICE

Always turn OFF the HVDC before the 120 VAC. Turning OFF 120 VAC power before HVDC power can result in equipment damage.

2. Turn OFF Axial Enable, HVDC, and 120VAC Service switches to disable Gantry motion.
3. Position OBC Chassis at 2 o'clock.

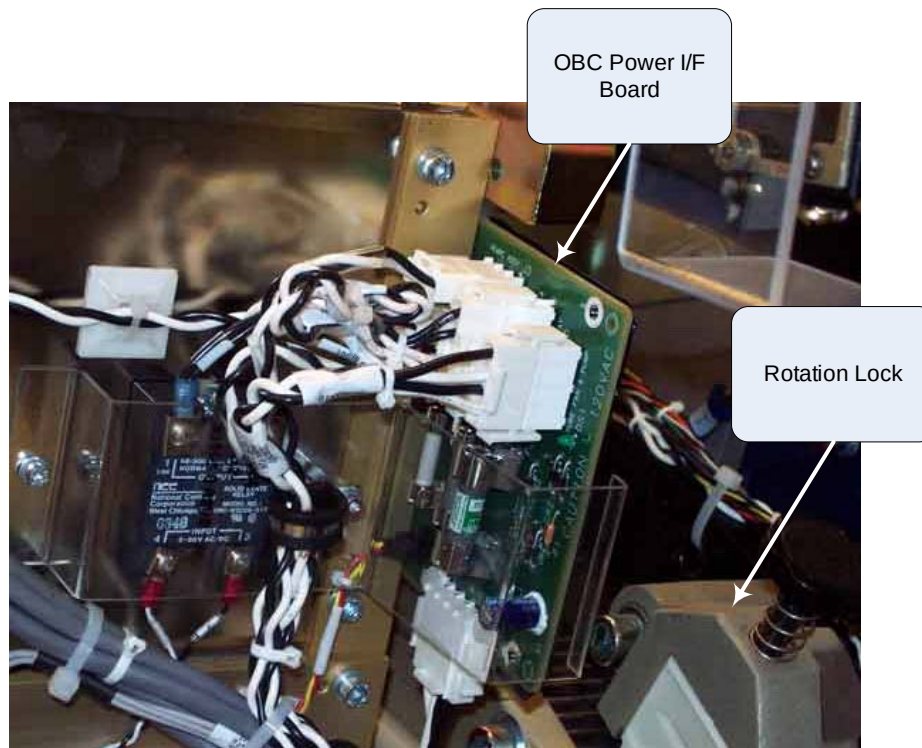


Figure 1: OBC Power I/F Board Location

With OBC Chassis at 2 O'clock position the OBC Power I/F Board will be located close to the Rotational Lock Assembly.

⚠ DANGER**ELECTROCUTION HAZARD.****HIGH VOLTAGE PRESENT.****USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE PERFORMING INSPECTION.**

4. Completely shutdown power at A1 panel, lock out and tag equipment.
5. Remove the Gantry Top and Back Covers to gain access to the Slip Ring Assembly. Follow cover removal procedures in Service Manual as needed.

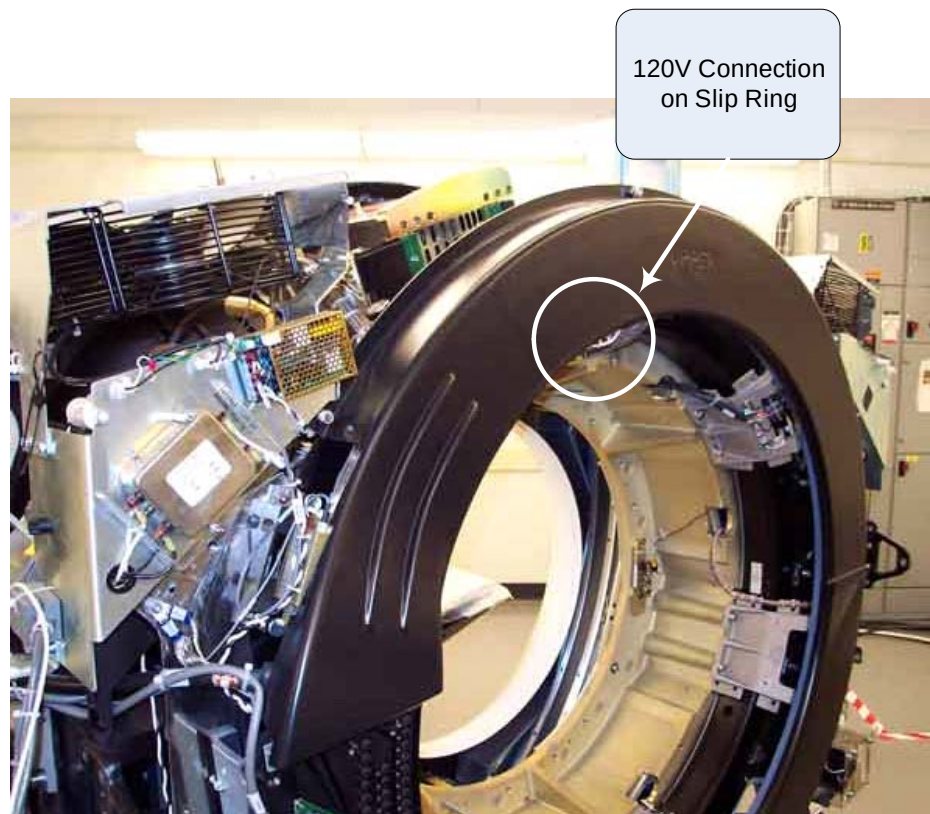


Figure 2: 120V Connection on Slip Ring Location

With OBC Chassis at 2 O'clock position the 120V Slip Ring connection will be located at the top left side of gantry viewed from the rear.

6. Remove the Upper Slip Ring Cover and set aside.

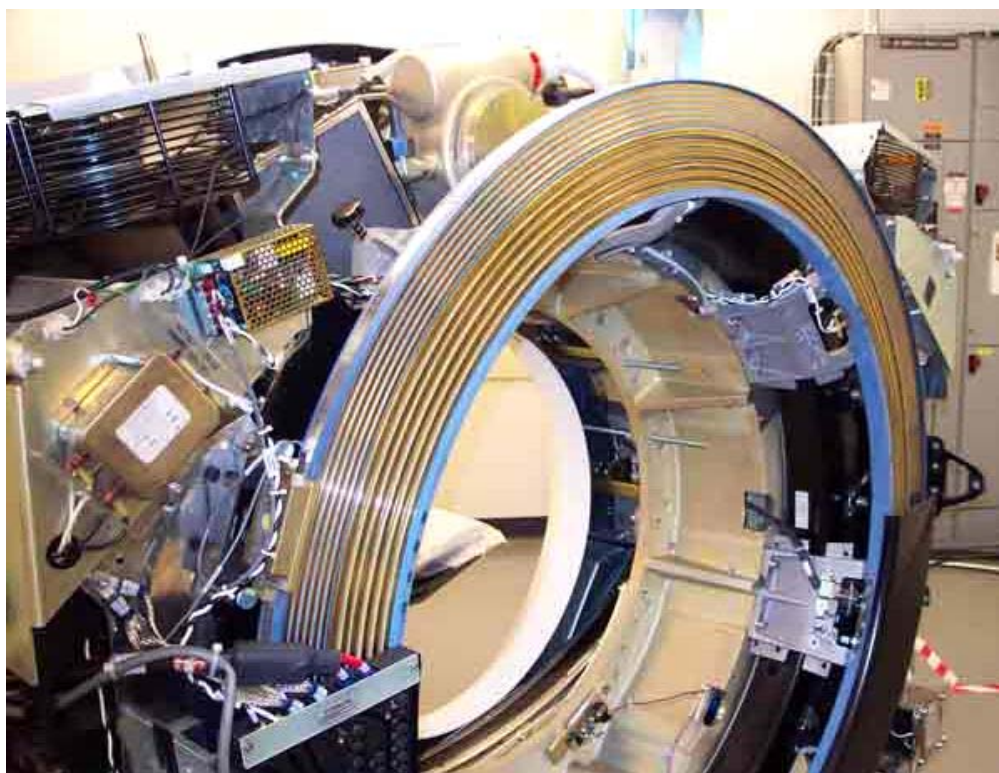


Figure 3: Upper Cover removed from Slip Ring

Inspect 120V Harness (2262943) Connections at Slip Ring

7. Inspect the 120Vac wiring connection on the Slip Ring and compare it to the Figure 4 below.

Note: It is critical to inspect both the Labeling on the Wires and the Connections!

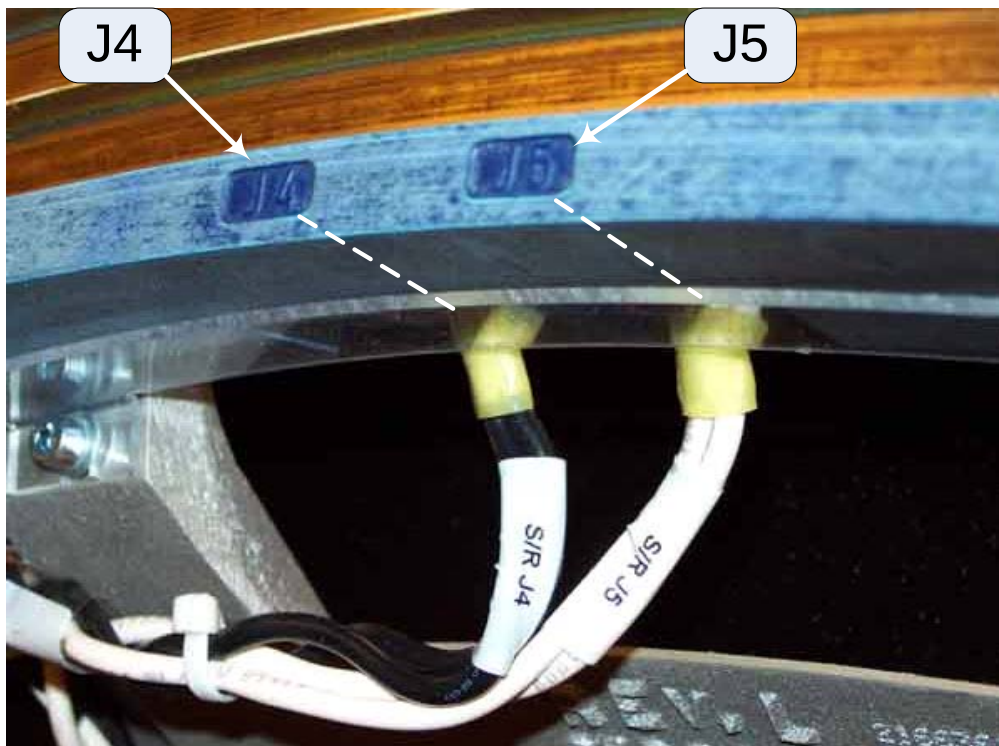


Figure 4: Proper wiring connection on Slip Ring

8. Verify the following wire labeling and connections:
 - a. Black Wire is labeled S/R J4 and is connected to the J4 connection on Slip Ring.
 - b. White Wire is labeled, S/R J5 and is connected to the J5 connection on Slip Ring.
9. If wires are not labeled correctly or are not connected as illustrated above, correct the wiring error and remove the old labels. Apply the new labels supplied with the kit, as indicated above.
10. Install Slip Ring Cover and Back Gantry Cover.

Electrical Check of 120Vac Harness

⚠ DANGER

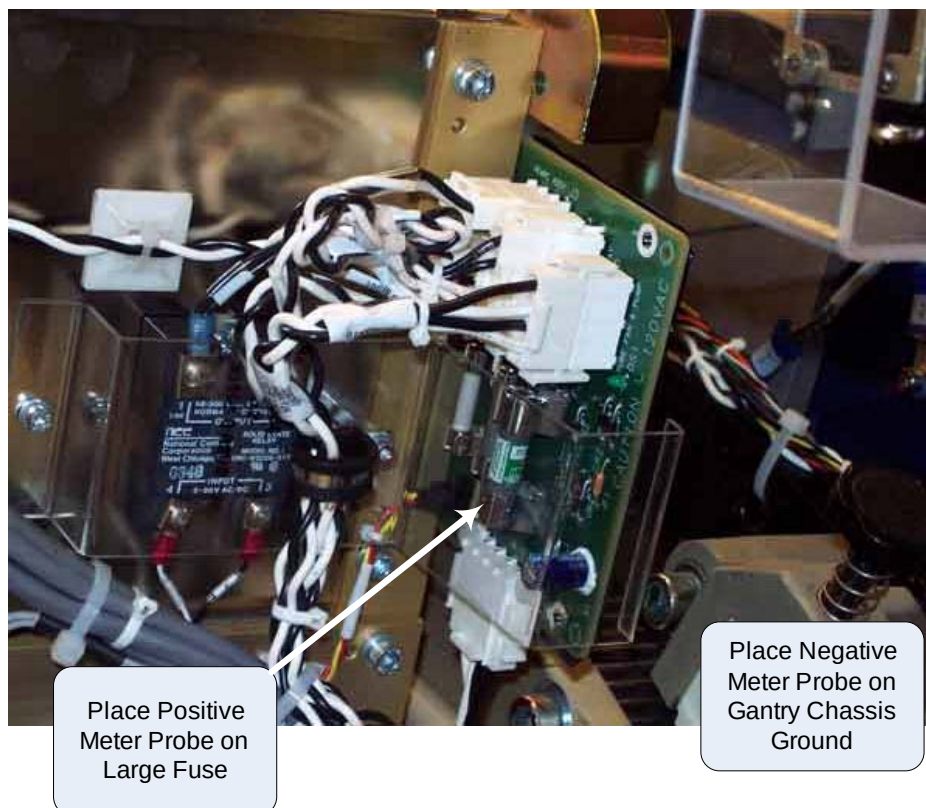


FLASH HAZARD, Hazard Class "0".

HIGH VOLTAGE PRESENT.

USE PROPER PPE PROCEDURES BEFORE PERFORMING ELECTRICAL CHECKS.

11. Remove Lock /Tag and turn on system power at A1 panel.
12. Turn ON 120VAC Service switch.
13. Using a Multimeter (DVM) set to measure AC Voltage, place the positive probe on the large fuse located on the OBC Power I/F Board and the negative probe on chassis ground.



Place Positive
Meter Probe on
Large Fuse

Place Negative
Meter Probe on
Gantry Chassis
Ground

Figure 5: Electrical Check for Proper Wiring

14. Verify that DVM reads 120VAC (100-130 VAC). If it does not measure 120VAC, ensure 120VAC is enabled or recheck wiring to Slip Ring. Troubleshoot as needed.
15. Install Gantry Top Covers.
16. Turn ON Axial Enable and HVDC Service switches to enable gantry rotation.

Finalization

17. Install the gantry side covers.
18. Perform a test scan to confirm system performance before returning system to the customer.
19. Close FMI Service Dispatch per local procedures.

NOTE:

If the gantry is found to be mis-wired and/or mis-labeled, correct the error and close the FMI Service Dispatch as a Code 1 (FMI Completed)

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